Machine Learning: A Quick Introduction

Machine learning (ML) is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every situation. An ML system learns from examples, then uses what it has learned to process new data.

Main Types of Machine Learning

1. Supervised learning: The model learns from labeled examples, where each input has a known answer. It is commonly used for classification and regression. For example, a model can predict house prices from features such as size, location, and number of rooms.

2. Unsupervised learning: The model looks for patterns in data without labeled answers. Clustering is a common technique. For example, a retailer may group customers by purchasing behavior.

3. Reinforcement learning: An agent learns by taking actions and receiving rewards or penalties. For example, an agent can learn a strategy for playing a game through repeated attempts.

Important Concepts

Features are the input variables used by a model. A **label** is the target answer in a supervised dataset. A **training set** is used to fit the model, while a **test set** evaluates how well it performs on data it has not seen during training.

Overfitting occurs when a model learns the training data too closely, including noise, and performs poorly on new examples. **Underfitting** occurs when a model is too simple to capture important patterns. Using separate training and test data helps reveal whether a model generalizes.

Common Algorithms

Linear regression predicts a continuous value, such as a temperature or price. Logistic regression is often used for classification, such as predicting whether an email is spam. Decision trees make predictions through a sequence of feature-based rules. K-means groups similar data points into a chosen number of clusters. The best algorithm depends on the task, the data, and the evaluation criteria.

Evaluating a Model

For classification, **accuracy** is the proportion of predictions that are correct. **Precision** measures how many predicted positive cases are actually positive, while **recall** measures how many actual positive cases the model finds. The F1-score combines precision and recall using their harmonic mean. For regression, metrics such as mean absolute error (MAE) measure the average absolute difference between predicted and actual values.

Quick Facts for RAG Testing

- Supervised learning uses labeled data.
- Unsupervised learning finds structure in unlabeled data.
- Overfitting usually harms performance on unseen data.
- K-means is a clustering algorithm.
- MAE is commonly used for regression evaluation.